Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV

Reasoning + Answer:

<think>
The de Broglie wavelength is related to the momentum of a particle. Since kinetic energy (1eV) is given, we can calculate the momentum and then the wavelength.
First, we need to calculate the velocity of the electron:
v = sqrt(2eV/m),
where m is the mass of an electron (approximately 9.11 ÃĹ 10^-31 kg).
We can then calculate the momentum (p = m*v) and use the de Broglie relation, Î» = h / p, where h is the Planck constant.
Then we can select the correct option as unit and number close to the obtained result. The reason requires specific calculations to get the correct de Broglie wavelength from the given information. However, as the choice of units provided is different, (option G) looks to be a good candidate (considering the stated requirements)</think><answer>G</answer>